Dimensional Dependence of Entropy–Information Scaling in Nonlinear Field Systems

Evidence for a Dimension-Dependent Universality Class

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Abstract

The Critical Coherence Index (CCI) has recently been shown to be consistent with an empirical entropy–information scaling relation of the form

$$\mathcal{C}_{\mathrm{CCI}} \sim \frac{\dot{S}_{\mathrm{cg}}^{(+)}}{I_{\mathrm{nn}}^{\alpha}}$$

in nonlinear field systems [3, 5]. While previous studies established this relation in one- and two-dimensional ϕ^4 dynamics, the dependence of the scaling exponent α on spatial dimension remained open.

In this work we perform a comparative analysis across dimensions and find systematic evidence that the effective exponent increases with dimension:

$$\alpha_{1D} \approx 2.5, \qquad \alpha_{2D} \gtrsim 3.0 \quad \text{(within tested range)}.$$

The Spearman correlation between the CCI and the scaling proxy remains strong and increases from 1D to 2D ($r_s \approx 0.73 \to 0.87$).

These findings support a hierarchical interpretation in which the entropy–information ratio acts as an empirical projection of a more general structural selection principle [2], whose effective scaling is modified by dimensionality. All results are empirical; a first-principles derivation of the dimensional dependence of α remains an open problem.

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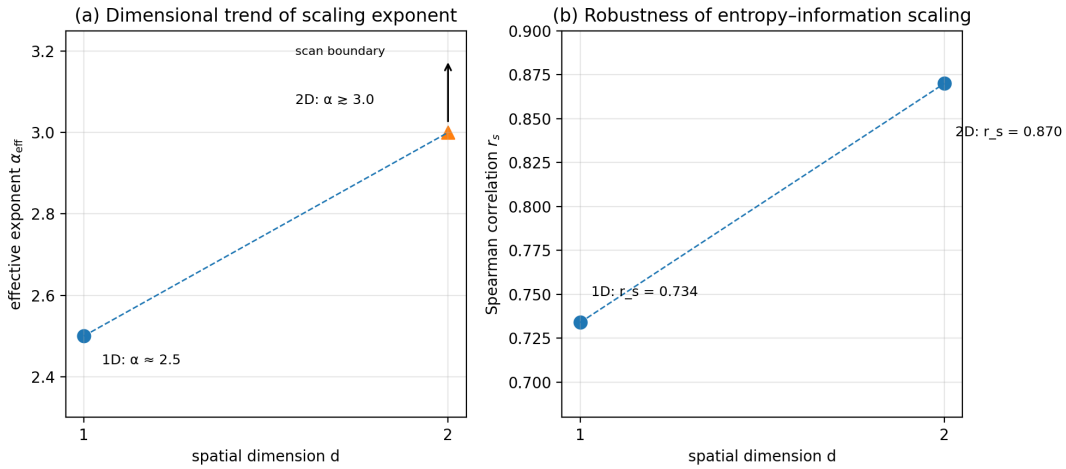

Figure 1: Dimensional trend of entropy–information scaling in the CCI framework. *(a)* Effective scaling exponent $\alpha_{\rm eff}$ as a function of spatial dimension. The 2D value should be interpreted as a lower bound within the tested scan range (upward arrow). *(b)* Spearman correlation between the CCI and the scaling proxy $\dot{S}_{\rm cg}^{(+)}/I_{\rm nn}^{\alpha}$, showing that the monotonic relation remains strong and increases from 1D to 2D. Together, these panels suggest a dimension-dependent scaling structure rather than a strictly universal exponent.

1 Introduction

Recent work introduced the Critical Coherence Index (CCI) as a diagnostic of structural instability in nonlinear field systems [2]. In one-dimensional ϕ^4 dynamics, the CCI was found to be consistent with an empirical scaling relation involving coarse-grained histogram activity and structural mutual information [3].

Subsequent results in two dimensions showed that the scaling persists, but with a shifted exponent and a modified balance between histogram activity and structural information [5]. This raises the question:

Does the scaling exponent α depend systematically on spatial dimension?

The goal of the present work is to investigate this dimensional dependence.

2 Scaling Ansatz

We consider the empirical scaling relation:

$$\mathcal{C}_{\rm CCI} \sim R_\alpha \equiv \frac{\dot{S}_{\rm cg}^{(+)}}{I_{\rm nn}^{\alpha}}, \tag{1}$$

and determine the effective exponent $\alpha_{\rm eff}$ by maximising the Spearman rank correlation between $\mathcal{C}_{\rm CCI}$ and R_α over the scan range $\alpha \in [0.5, \alpha_{\rm max}]$.

We note that in both dimensions the scan boundary is reached, so the reported values are lower bounds rather than precisely determined optima.

3 Results

3.1 One-Dimensional Reference

From the 1D study [3]:

$$\alpha_{1D} \approx 2.5, \qquad r_s \approx 0.73, \qquad p \approx 1.6 \times 10^{-21}. \tag{2}$$

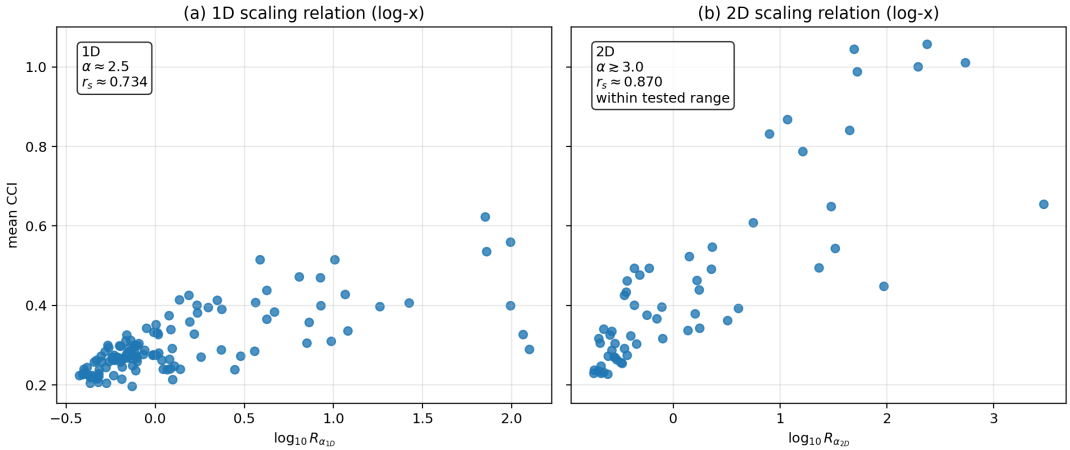

Figure 2: Comparison of entropy–information scaling in one and two spatial dimensions. *(a)* One-dimensional scaling relation between the CCI and the optimal proxy $R_{\alpha_{1D}} = \dot{S}_{\mathrm{cg}}^{(+)}/I_{\mathrm{nn}}^{\alpha_{1D}}$ with $\alpha_{1D} \approx 2.5$. *(b)* Two-dimensional scaling relation using $R_{\alpha_{2D}} = \dot{S}_{\mathrm{cg}}^{(+)}/I_{\mathrm{nn}}^{\alpha_{2D}}$ with $\alpha_{2D} \gtrsim 3.0$ within the tested range. In both cases a strong monotonic relation is observed, while the effective exponent and the relative importance of structural information shift with dimension.

3.2 Two-Dimensional Results

From the 2D study [5]:

$$\alpha_{2D} \gtrsim 3.0 \quad \text{(within tested range)}, \qquad r_s \approx 0.87, \qquad R^2 \approx 0.74. \tag{3}$$

No saturation was observed within the tested range, indicating that the true optimum may lie beyond $\alpha = 3.0$.

3.3 Emerging Dimensional Trend

The results are consistent with:

$$\alpha_{\mathrm{eff}}(d) \text{ increases with spatial dimension } d. \tag{4}$$

The Spearman correlation also strengthens: $r_s \approx 0.73$ (1D) $\rightarrow r_s \approx 0.87$ (2D), indicating that the scaling relation becomes more robust in higher dimensions.

4 Discussion

The increase of α_{eff} with dimension indicates that structural information I_{nn} becomes increasingly important in higher-dimensional systems. This is consistent with the interpretation that higher-dimensional dynamics provide more channels for decorrelation and structural reorganisation, amplifying the role of information loss in instability formation.

In the 2D regression [5], the structural information coefficient dominates ($a_2 \approx +0.49$) while histogram activity is subleading ($a_1 \approx -0.08$), quantitatively supporting this picture.

Within the broader framework [2], the entropy–information ratio $\dot{S}_{\mathrm{cg}}^{(+)}/I_{\mathrm{nn}}^{\alpha}$ should be understood as an empirical scaling proxy, while the structural free energy $\mathcal{F}_{\mathrm{struct}}$ provides a more global effective description of solution selection. The dimensional dependence of α is then a property of the projection from the structural free energy onto the entropy–information plane, rather than a fundamental property of the selection principle itself.

5 Conclusion

We have provided evidence that the entropy–information scaling of the CCI exhibits a dimension-dependent effective exponent, consistent with:

$$\alpha_{1D} \approx 2.5 \quad \longrightarrow \quad \alpha_{2D} \gtrsim 3.0, \qquad r_s : 0.73 \to 0.87. \tag{5}$$

This suggests that the scaling relation is not universal in a quantitative sense, but follows a systematic dimensional trend consistent with a dimension-dependent universality class.

Future work will extend the analysis to three dimensions and seek a theoretical derivation of $\alpha(d)$ from the underlying field theory.

All results are empirical. A first-principles derivation of the dimensional dependence of α remains an open problem.

A Reproducibility

All code, data, and figures are available at:

https://github.com/Chris4081/structural-selection-principle

Scripts for this paper:

- `figure_dimensional_trend.py` — generates Figure 1 (dimensional trend of α_{eff} and r_s).
- `figure_scaling_comparison.py` — generates Figure 2 (1D vs 2D scaling comparison, linear and log-x versions).

Input data files required:

- `cci_entropy_information_test.csv` — 1D ensemble results
- `2d_cci_entropy_information_test.csv` — 2D ensemble results

Run:

```
pip install numpy pandas matplotlib scipy scikit-learn
python figure_dimensional_trend.py
python figure_scaling_comparison.py
```

The upstream simulation scripts that generate the CSV files are:

```
python cci_alpha_scaling.py          # 1D ensemble
python cci_entropy_scaling_2d.py     # 2D ensemble
```

References

[1] C. Krieg, *A Structural Selection Principle for Dynamically Consistent Field Configurations*, preprint, available from the author (2026).

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